The coordinate ring of the k-fold iterated commutator locus for 2×2 matrices, via a signed encoding[1]

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Abstract

This is a companion to *ring-of-k-commuting-2x2-matrices.typ*, proving the **same** four results about the coordinate ring of the k-fold iterated commutator locus for 2×2 matrices – codimension 2, exactly 3 minimal generators, Cohen-Macaulayness, and radicality, assembling into the same Primary Decomposition Theorem $I_k = P_2 \cap \cdots \cap P_k$ – using a different encoding of a traceless matrix as a triple in $\mathbb{F}^3$. Where the canonical article's encoding turns the bracket into a cross product composed with the twist $\tau(a,b,c) = (b,a,2c)$ (a coordinate swap **and** a scaling by 2), this one turns it into a cross product composed with $D := \mathrm{diag}(-1,-1,1)$ – no swap, no scaling beyond a sign, and $D^2 = I$ (an involution, unlike τ). The reason a purely-sign twist exists at all is structural, not luck: the invariant quadratic form of $\mathfrak{sl}_2$ in these coordinates – the matrix determinant itself, $\det(A_0) = -x^2 - yz$ for traceless A_0 – has an indefinite signature that Sylvester's law of inertia says D's sign pattern already realizes optimally over any field of characteristic $\neq 2$. Every theorem below transfers from the canonical article for free (any invertible linear re-encoding gives an isomorphic ideal, by the classical $\mathrm{GL}(3)$-transformation law for the cross product), so only the exposition, not the mathematics, is new here.

1. Introduction

This article studies the **same object** as its companion *ring-of-k-commuting-2x2-matrices.typ*: for 2×2 matrices $A_1, \ldots, A_k$, the left-normed iterated commutator $[A_1, \ldots, A_k] := [\ldots[[A_1, A_2], A_3], \ldots, A_k]$, and I_k, the ideal in the $3k$-variable reduced-coordinate polynomial ring R_k cutting out its vanishing locus. The companion article works entirely with the encoding $\mathrm{red}(A) := (a_{12}, a_{21}, a_{11} - a_{22})$ and the twist $\tau(a,b,c) := (b,a,2c)$ needed to turn $[A,B]$'s bracket into a cross product; this article uses instead

$$\begin{aligned} \mathrm{enc}(A) := (&a_{11} - a_{22}, \\ &a_{12} + a_{21}, \\ &a_{12} - a_{21}) \in \mathbb{F}^3, \end{aligned} \tag{1}$$

which turns out to need only the sign-diagonal twist $D := \mathrm{diag}(-1,-1,1)$ in the analogous identity (Theorem 1 below) – no coordinate swap, and only a sign rather than a scaling by 2. Both encodings are legitimate; the point of writing this second article is that the resulting exposition is genuinely simpler, not merely different, for reasons made precise in Remark 2 below. Every result proven in the companion article – codimension, generator count, Cohen-Macaulayness, radicality, and the resulting Primary Decomposition Theorem – holds here too, by the **same** proofs in substance; what changes is Section 3′s derivation, the concrete low-degree formulas displayed in examples throughout, and the Jacobian-recursion machinery of §7 (which needs D in place of τ). See the companion article's own Introduction for the classical-commuting-variety backstory (Motzkin-Taussky, Gerstenhaber, the Artin-Hochster conjecture) – not repeated here to avoid duplicating material unrelated to the encoding question this article is actually about.

2. Notation

Notation used across more than one later result; see the companion article's own Notation section for anything not listed here (the only genuinely new symbol below is D, replacing τ).

Symbol	Meaning	Defined in

[1]Working draft, 2026-08-19. A **companion** to *ring-of-k-commuting-2x2-matrices.typ* (kept exactly as it was), encoding a traceless matrix's free entries differently – see *working-notes/alternative-encodings-musings.org* for the derivation and the comparison against two other candidates, and this footnote's closing paragraph for why the difference is more than cosmetic. Every theorem statement below is verbatim identical to the canonical article's; only the Setup section, the explicit low-degree examples, and the Jacobian-recursion machinery of §7 needed to be redone. Uses "theorion" for theorem environments, matching the canonical article's own standing default.

$\overline{v}$ (also $\overline{v}_i$; components $v_{i,1}, v_{i,2}, v_{i,3}$, or $v_{i,j}$)	reduced-coordinate vector in $\mathbb{F}^3$, and its blocks	§3, Setup
D	the sign twist $D := \mathrm{diag}(-1,-1,1)$	§3, Setup (D as a matrix throughout)
$\mathbb{F}$	ambient field, characteristic $\neq 2$	§3, after Definition 3's example
R_k	the polynomial ring $\mathbb{F}[\overline{v}_1, \ldots, \overline{v}_k]$	Definition 3
I_k	the defining ideal, $I_k \subset R_k$	Definition 3
$\overline{w}_j$	(a) the recursive vector in $\mathbb{F}^3$; (b) as a polynomial map $\mathbb{A}^{3j} \to \mathbb{A}^3$	(a) Definition 3; (b) Lemma 7
U_j	the Zariski-open locus where $\overline{w}_j \neq 0$	Lemma 7
π	the projection $\mathbb{A}^{3k} \to \mathbb{A}^{3(k-1)}$ forgetting $\overline{v}_k$	Definition 11
Φ	the map $\mathbb{A}^{3(k-1)} \times \mathbb{A}^1 \to \mathbb{A}^{3k}$	Definition 11
Y_1, Y_2	the two strata of $V(I_k)$	Definition 11
P_j $(j = 2, \ldots, k)$	the primary components of I_k	Definition 20
J_k	the Jacobian of $\overline{w}_k$	Lemma 24
$[u]_\times$	skew matrix with $[u]_\times v = u \times v$	Lemma 24

3. Setup: the signed encoding and the defining ideal

Step 1: discard the scalar part. Identical to the companion article: $[A, B] = [A_0, B_0]$ for A_0, B_0 the traceless remainders, since scalar matrices are central.

Step 2: encode the traceless remainder. A traceless matrix $\left(\begin{smallmatrix} x & y \\ z & -x \end{smallmatrix}\right)$ has three free entries. Rather than reading them off directly (x, y, z themselves – shown in Remark 2 below to be a strictly **worse** choice, needing two separate factors of 2), take

$$\begin{aligned} \mathrm{enc}(A) := (&a_{11} - a_{22}, \\ &a_{12} + a_{21}, \\ &a_{12} - a_{21}) \in \mathbb{F}^3, \end{aligned} \tag{2}$$

the traceless $(1,1)$-entry doubled (as in the companion article's d_a), and A's two off-diagonal entries replaced by their sum and difference. As with the companion article's red, no division is needed to **compute** enc – only its inverse (recovering a_{12}, a_{21} individually as $\frac{v_2+v_3}{2}$ and $\frac{v_2-v_3}{2}$) needs 2 invertible, exactly the same hypothesis the companion article already carries, not a new one.

Step 3: form the commutator. As before, $[A_0, B_0]$ is again traceless, hence again encodable.

Step 4: the encoded bracket is the cross product, up to one fixed sign twist.

Theorem 1 (the basic identity)

For traceless 2×2 matrices A_0, B_0 over $\mathbb{F}$ (char $\mathbb{F} \neq 2$),

$$\mathrm{enc}([A_0, B_0]) = D(\mathrm{enc}(A_0) \times \mathrm{enc}(B_0)), \quad D := \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{3}$$

Proof of Theorem 1. A direct computation, carried out symbolically over generic entries $a_{11}, \ldots, b_{22}$ (not merely checked at sampled points) in *code/verify_lorentzian_encoding_identity.sage*: both sides, expanded in the 8 free entries, agree term by term. □

Remark 2 (why this particular twist, and not some other one)

Three concrete respects in which D improves on the companion article's $\tau(a,b,c) = (b,a,2c)$, each checked, not merely asserted:

1. ***D* does not mix coordinates.** τ swaps two of its three inputs; D acts diagonally, one coordinate at a time. Consequence: the companion article needs a page-length remark ("why $\overline{w}_j$ carries no outer τ") to justify that rescaling+swapping a generating set by an invertible linear map doesn't change the ideal generated. Here the analogous fact is immediate: **individually** negating two of three generators obviously leaves the ideal unchanged, generator by generator – no swap to account for. (Still verified computationally, not merely asserted: $\mathrm{ideal}(\overline{w}_2) = \mathrm{ideal}(D\overline{w}_2)$, confirmed by direct Gröbner-basis equality in *code/verify_encoding_gl3_equivariance.sage*.)
2. ***D* is an involution; *τ* is not.** $D^2 = I$ exactly (immediate from the diagonal entries being ± 1), while $\tau^2 = \mathrm{diag}(1,1,4) \neq I$.
3. ***D* is not an arbitrary matrix that happens to work – it is (up to a constant) the matrix of $\mathfrak{sl}_2$'s own Killing form, in these coordinates.** For traceless $A_0 = \left(\begin{smallmatrix} x & y \\ z & -x \end{smallmatrix}\right)$, $\det(A_0) = -x^2 - yz$, an **indefinite** quadratic form (never definite: $-yz = -((y+z)/2)^2 + ((y-z)/2)^2$ is a genuine difference of squares whenever $2 \neq 0$). Checked directly: $-\mathrm{enc}(A)_1^2 - \mathrm{enc}(A)_2^2 + \mathrm{enc}(A)_3^2 = 4\det(A_0)$, exactly (Sage, symbolic). So D's sign pattern **is** det's own signature, read off in these coordinates – not a bookkeeping accident. (Lie-theoretically: $\mathfrak{sl}_2(\mathbb{R}) \cong \mathfrak{so}(2,1)$, the Lorentz algebra, not the compact $\mathfrak{so}(3)$ one gets only after adjoining i – classical, not cited to a specific source here, the same way the companion article leaves certain folklore facts uncited rather than inventing a citation.) By Sylvester's law of inertia the signature of a quadratic form is an invariant of the form itself, not of the coordinates chosen to express it – so no linear re-encoding over a field where -1 is not a square of 1 (in particular no real or rational one) can ever remove this sign entirely; D's single-sign twist is, in this precise sense, already the simplest one achievable, not merely simpler than τ by luck.

See *working-notes/alternative-encodings-musings.org* for the fuller discussion, including why the "obvious" un-permuted reading of the matrix entries is actually **worse** than either encoding (needs a swap **and** two separate factors of 2), and a remark on why the field-extension route to a literal, untwisted cross product ($i \in \mathbb{F}$, giving $\mathfrak{su}(2) \cong \mathfrak{so}(3)$ instead) is a real alternative but a strictly narrower one.

◇

Definition 3 (reduced iterated commutator, signed encoding)

$$\overline{w}_2 := \overline{v}_1 \times \overline{v}_2, \quad \overline{w}_j := D\big(\overline{w}_{j-1}\big) \times \overline{v}_j \quad (j = 3, \ldots, k). \tag{4}$$

Let $R_k := \mathbb{Q}[\overline{v}_1, \ldots, \overline{v}_k]$ ($\overline{v}_i \in \mathbb{Q}^3$, $3k$ variables) and $I_k \subset R_k$ the ideal generated by the three components of $\overline{w}_k$.

♣

As in the companion article, "$R_k := \mathbb{Q}[\overline{v}_1, \ldots, \overline{v}_k]$" is shorthand for the polynomial ring in $3k$ scalar variables $v_{i,j}$, grouped into k blocks of three – see that article's own worked-out R_2 display, identical here up to variable naming.

Example 4 ($\overline{w}_k$ and I_k, explicitly, for $k = 2, 3, 4$)

Computed directly (*code/compute_signed_encoding_examples.sage*), not hand-derived. At $k = 2$, the three components of $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$:

$$v_{1,2}v_{2,3} - v_{1,3}v_{2,2}, \quad v_{1,3}v_{2,1} - v_{1,1}v_{2,3}, \quad v_{1,1}v_{2,2} - v_{1,2}v_{2,1}. \tag{5}$$

(Identical to the companion article's $k = 2$ display – the two encodings agree exactly at the very first stage, since neither τ nor D is applied yet.) At $k = 3$, the three components of $\overline{w}_3 = D(\overline{w}_2) \times \overline{v}_3$ – now with **every coefficient equal to ± 1**, no factor of 2 anywhere, unlike the companion article's τ-twisted $\overline{w}_3$:

$$v_{1,2}v_{2,1}v_{3,2} - v_{1,1}v_{2,2}v_{3,2} - v_{1,3}v_{2,1}v_{3,3} + v_{1,1}v_{2,3}v_{3,3}, \tag{6}$$

$$-v_{1,2}v_{2,1}v_{3,1} + v_{1,1}v_{2,2}v_{3,1} - v_{1,3}v_{2,2}v_{3,3} + v_{1,2}v_{2,3}v_{3,3}, \tag{7}$$

$$v_{1,3}v_{2,1}v_{3,1} - v_{1,1}v_{2,3}v_{3,1} + v_{1,3}v_{2,2}v_{3,2} - v_{1,2}v_{2,3}v_{3,2}, \tag{8}$$

and $I_3 := \langle w_{3,1}, w_{3,2}, w_{3,3} \rangle$. At $k = 4$, $\overline{w}_4 = D(\overline{w}_3) \times \overline{v}_4$: again every coefficient is ± 1; the first component of $\overline{w}_4$ is

$$-v_{1,3}v_{2,1}v_{3,1}v_{4,2} + v_{1,1}v_{2,3}v_{3,1}v_{4,2} - v_{1,3}v_{2,2}v_{3,2}v_{4,2} + v_{1,2}v_{2,3}v_{3,2}v_{4,2} \tag{9}$$

$$+v_{1,2}v_{2,1}v_{3,1}v_{4,3} - v_{1,1}v_{2,2}v_{3,1}v_{4,3} + v_{1,3}v_{2,2}v_{3,3}v_{4,3} - v_{1,2}v_{2,3}v_{3,3}v_{4,3}, \tag{10}$$

the other two of the same shape (omitted for space; all three components of $\overline{w}_4$, and $I_4 := \langle w_{4,1}, w_{4,2}, w_{4,3} \rangle$, in *code/compute_signed_encoding_examples.sage*'s output).

Stated over $\mathbb{Q}$, as in the companion article, but every result below holds over an arbitrary field $\mathbb{F}$ of characteristic $\neq 2$ (needed only for enc's own invertibility as a map on matrices – see Step 2 above; D itself is invertible, in fact $\det D = 1$, in **every** characteristic, though this article does not pursue whether the ring-theoretic results below could be extended to characteristic 2 by some other route – flagged as a genuinely open question, not investigated here, prompted purely by the observation that every formula in Example 4 above is already free of any numeric coefficient besides ± 1).

4. The determinantal structure and the generator count

Theorem 5 (determinantal structure)

For $k \geq 3$, I_k is exactly the ideal of 2×2 minors of the 2×3 matrix

$$\begin{pmatrix} D(\overline{w}_{k-1}) \\ \overline{v}_k \end{pmatrix}, \tag{11}$$

i.e. $V(I_k)$ is the preimage, under $(\overline{v}_1, \ldots, \overline{v}_k) \mapsto \begin{pmatrix} D(\overline{w}_{k-1}(\overline{v}_1, \ldots, \overline{v}_{k-1})) \\ \overline{v}_k \end{pmatrix}$, of the classical rank-$\leq 1$ determinantal variety of 2×3 matrices. (At $k = 2$ the same statement holds with $D(\overline{w}_{k-1})$ replaced by $\overline{v}_1$ itself.)

Proof of Theorem 5. Immediate from Definition 3's recursion and the fact that a cross product's three components are, up to sign, exactly the 2×2 minors of the 2×3 matrix with the two crossed vectors as rows – and checked independently at $k = 2, \ldots, 4$ against Theorem 1's identity, chained (*code/compute_signed_encoding_examples.sage*); the underlying polynomial identity makes this a two-line induction valid for every k, exactly as in the companion article's own Theorem 5. □

Example 6 (the 2×3 matrix of Theorem 5, for $k = 3, 4$)

At $k = 3$, the first row is $D(\overline{w}_2) = (-w_{2,1}, -w_{2,2}, w_{2,3})$ – note this is **literally** $\overline{w}_2$ with its first two components negated, not a mix of $\overline{w}_2$'s components the way the companion article's $\tau(\overline{w}_2)$ swaps two of them:

$$\begin{pmatrix} D(\overline{w}_2) \\ \overline{v}_3 \end{pmatrix} = \begin{pmatrix} -v_{1,2}v_{2,3} + v_{1,3}v_{2,2} & -v_{1,3}v_{2,1} + v_{1,1}v_{2,3} & v_{1,1}v_{2,2} - v_{1,2}v_{2,1} \\ v_{3,1} & v_{3,2} & v_{3,3} \end{pmatrix}. \tag{12}$$

At $k = 4$, the first row is $D(\overline{w}_3)$, entries cubic, every coefficient ± 1:

$$\begin{pmatrix} -v_{1,2}v_{2,1}v_{3,2} + v_{1,1}v_{2,2}v_{3,2} + v_{1,3}v_{2,1}v_{3,3} - v_{1,1}v_{2,3}v_{3,3} & \cdots & \cdots \\ v_{4,1} & v_{4,2} & v_{4,3} \end{pmatrix} \tag{13}$$

(other two entries of the same shape, omitted for width – all displayed in full in *code/compute_signed_encoding_examples.sage*).

◇

Lemma 7 (non-degeneracy)

For every $j \geq 2$, the polynomial map $\overline{w}_j : \mathbb{A}^{3j} \to \mathbb{A}^3$ is not identically zero. Hence $U_j := \{p \in \mathbb{A}^{3j} : \overline{w}_j(p) \neq 0\}$ is a nonempty Zariski-open subset.

♡

Proof of Lemma 7. By induction: $\overline{w}_2(e_1, e_2) = e_1 \times e_2 = e_3 \neq 0$; given $c := \overline{w}_{j-1}(p) \neq 0$, $D(c) \neq 0$ since D is invertible ($\det D = 1 \neq 0$, in **every** characteristic – a strictly weaker requirement than the companion article's own use of τ's invertibility, which needs characteristic $\neq 2$ specifically), so at least one standard basis vector is not proportional to $D(c)$, and a cross product vanishes only on linearly dependent pairs, giving $\overline{w}_{j(p,e_i)} = D(c) \times e_i \neq 0$ for a suitable e_i. □

Lemma 8 (independent same-degree generators are minimal)

If a homogeneous ideal in a graded ring is generated by finitely many forms of the same degree that are linearly independent over the base field, that generating set is minimal.

♡

Proof of Lemma 8. A polynomial-coefficient combination reproducing one of the generators from the others, while staying in the same degree, can only use degree-0 (constant) coefficients – so a redundant generator would violate linear independence directly. □

Lemma 9 (the three generators of I_k are independent)

The three components of $\overline{w}_k$ are linearly independent over $\mathbb{F}$, for every $k \geq 2$.

♡

Proof of Lemma 9. Checked directly (coefficient-matrix rank 3) at $k = 2, 3, 4$. Structurally: a nontrivial constant linear relation among the components of $\overline{w}_k = D(\overline{w}_{k-1}) \times \overline{v}_k$ would force $D(\overline{w}_{k-1})$ itself to be a constant vector, which it is not for $k \geq 3$ (Lemma 7's non-degeneracy, D invertible) nor for $k = 2$ ($\overline{v}_1$ is a free variable block). □

Theorem 10 (I_k has exactly 3 minimal generators)

For every $k \geq 2$: I_k has exactly 3 minimal generators.

♡

Proof of Theorem 10. Immediate from Lemma 8 applied to Lemma 9's independence. □

5. The geometric structure of $V(I_k)$

This section's topological lemma (continuous images and closures preserve irreducibility) and the remark on why the closure, not the raw image, is what Theorem 16's proof needs, are identical to the companion article's own §5 material and not repeated here – neither uses τ or D at all, so nothing changes.

Definition 11 (the strata Y_1, Y_2)

For $k \geq 3$, write π for the projection forgetting $\overline{v}_k$, and Φ for

$$\Phi : \mathbb{A}^{3(k-1)} \times \mathbb{A}^1 \to \mathbb{A}^{3k}, \quad \Phi(p, \lambda) := (p, \lambda D(\overline{w}_{k-1}(p))). \tag{14}$$

$$Y_1 := V(I_{k-1}) \times \mathbb{A}^3, \quad Y_2 := \overline{\Phi(\mathbb{A}^{3k-2})}. \tag{15}$$

At $k = 2$, $Y_2 := V(I_2)$ directly, matching Theorem 5′s base case.

Lemma 12 (Y_2 is irreducible)

For $k \geq 3$: Y_2 is irreducible, and its defining ideal (the graph ideal of Φ with λ eliminated) is prime.

Proof of Lemma 12. Identical argument to the companion article's own Lemma 12: Φ is a polynomial map, hence Zariski-continuous, with irreducible domain, so its image and closure are irreducible; the elimination ideal is the kernel of a map into a domain, hence prime. Additionally corroborated computationally at $k = 3, 4$ by computing the elimination ideal directly (below) and confirming primality in Sage. □

Example 13 (the defining ideal of Y_2, at $k = 3, 4$)

Computed directly by eliminating λ from the graph ideal of Φ (*code/compute_signed_encoding_examples.sage*). At $k = 3$:

$$v_{2,1}v_{3,1} + v_{2,2}v_{3,2} - v_{2,3}v_{3,3}, \quad v_{1,1}v_{3,1} + v_{1,2}v_{3,2} - v_{1,3}v_{3,3}, \tag{16}$$

$$v_{1,2}v_{2,1}v_{3,2} - v_{1,1}v_{2,2}v_{3,2} - v_{1,3}v_{2,1}v_{3,3} + v_{1,1}v_{2,3}v_{3,3} \tag{17}$$

– again every coefficient ± 1, in contrast with the companion article's $k = 3$ display, which has explicit factors of 2 on two of the three generators (an artifact of τ's own doubling, not present here). This ideal is a proper factor of I_3, matching P_3 in the component-count example below. At $k = 4$:

$$v_{3,1}v_{4,1} + v_{3,2}v_{4,2} - v_{3,3}v_{4,3}, \tag{18}$$

$$v_{1,3}v_{2,2}v_{4,1} - v_{1,2}v_{2,3}v_{4,1} - v_{1,3}v_{2,1}v_{4,2} + v_{1,1}v_{2,3}v_{4,2} + v_{1,2}v_{2,1}v_{4,3} - v_{1,1}v_{2,2}v_{4,3}, \tag{19}$$

$$v_{1,3}v_{2,1}v_{3,1}v_{4,2} - v_{1,1}v_{2,3}v_{3,1}v_{4,2} + v_{1,3}v_{2,2}v_{3,2}v_{4,2} - v_{1,2}v_{2,3}v_{3,2}v_{4,2} \tag{20}$$

$$-v_{1,2}v_{2,1}v_{3,1}v_{4,3} + v_{1,1}v_{2,2}v_{3,1}v_{4,3} - v_{1,3}v_{2,2}v_{3,3}v_{4,3} + v_{1,2}v_{2,3}v_{3,3}v_{4,3}. \tag{21}$$

Example 14 (a suitable point of Y_2, concretely, at $k = 3$)

Take $\overline{v}_1 = (1, 0, 0)$, $\overline{v}_2 = (0, 1, 0)$ (so $\overline{v}_1 \in U_2$, Lemma 7), giving $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2 = (0, 0, 1)$ and $D(\overline{w}_2) = (0, 0, 1)$ – here D acts as the identity, since both of $\overline{w}_2$'s sign-flipped coordinates already vanish, a strictly cleaner example than the companion article's own D-analogue point ($\tau(\overline{w}_2) = (0, 0, 2)$ there, carrying an explicit factor of 2). Taking $\lambda = 1$, i.e. $\overline{v}_3 := D(\overline{w}_2) = (0, 0, 1)$, gives

$$(\overline{v}_1, \overline{v}_2, \overline{v}_3) = ((1, 0, 0), (0, 1, 0), (0, 0, 1)) \in Y_2 \subset \mathbb{A}^9. \tag{22}$$

Directly: $\overline{w}_3 = D(\overline{w}_2) \times \overline{v}_3 = (0, 0, 1) \times (0, 0, 1) = 0$, confirming this point lies in $V(I_3)$.

Lemma 15 (dimension of Y_2, and where it lives)

With Y_2 as in Lemma 12 and U_{k-1} as in Lemma 7: $\dim Y_2 = 3k - 2$, and Y_2 meets $\pi^{-1}(U_{k-1})$ nonemptily.

Proof of Lemma 15. Over U_{k-1}, Φ is injective: p is recovered directly, and λ by dividing any nonzero coordinate of $\overline{v}_k = \lambda D(\overline{w}_{k-1}(p))$ by the corresponding coordinate of $D(\overline{w}_{k-1}(p))$ – a division by a nonzero **field element**, needing nothing about characteristic beyond that coordinate being nonzero, which $D(\overline{w}_{k-1}(p)) \neq 0$ guarantees on U_{k-1} (D invertible). Injectivity gives $\dim Y_2 = \dim(U_{k-1} \times \mathbb{A}^1) = 3k - 2$; since U_{k-1} is nonempty (Lemma 7), $\Phi(U_{k-1} \times \{1\}) \subseteq Y_2 \cap \pi^{-1}(U_{k-1})$ is nonempty. □

Theorem 16 (I_k has codimension 2)

For every $k \geq 2$: $\dim V(I_k) = 3k - 2$, i.e. $\operatorname{codim}(I_k) = 2$.

Proof of Theorem 16. Induction on k, identical in structure to the companion article's Theorem 16: base case $k = 2$ is the standard dimension of a generic rank-≤ 1 locus of a 2×3 matrix [3] inductive step splits $V(I_k) = Y_1 \cup Y_2$ by whether $D(\overline{w}_{k-1}) = 0$ or $\overline{v}_k$ is proportional to it (a direct case split, needing no closure argument, exactly as before), giving $\dim Y_1 = 3k - 2$ by the inductive hypothesis and $\dim Y_2 = 3k - 2$ by Lemma 15. □

Proposition 17 ($Y_2 \not\subseteq Y_1$)

With Y_1, Y_2 as above: $Y_2 \not\subseteq Y_1$.

Proof of Proposition 17. Pick $p \in U_{k-1}$; $\Phi(p, 0) = (p, 0) \in Y_2$ but $\overline{w}_{k-1}(p) \neq 0$ puts $(p, 0) \notin Y_1$. □

Proposition 18 (no component of Y_1 lies in Y_2)

No irreducible component Z of $V(I_{k-1})$ has $Z \times \mathbb{A}^3 \subseteq Y_2$.

Proof of Proposition 18. Both irreducible, same dimension $3k - 2$ (Theorem 16, Lemma 15), so containment forces equality; refuted since $Z \times \mathbb{A}^3$ is disjoint from $\pi^{-1}(U_{k-1})$ ($\overline{w}_{k-1}$ vanishes on Z) while Y_2 meets it (Lemma 15). □

Theorem 19 (component count)

For every $k \geq 2$: $V(I_k)$ has exactly $k - 1$ irreducible components, each of dimension $3k - 2$.

Proof of Theorem 19. Induction on k: base case $k = 2$ is the irreducibility of the rank-≤ 1 locus of a generic 2×3 matrix [3] step combines the inductive hypothesis's $k - 2$ components of Y_1 with Y_2 itself, no containments among them by Proposition 17/Proposition 18, giving $(k - 2) + 1 = k - 1$. □

Definition 20 (the components P_j)

As in the companion article: P_j ($j = 2, \ldots, k$) is Y_2's own defining ideal run at parameter j, viewed inside R_k by leaving $\overline{v}_{j+1}, \ldots, \overline{v}_k$ free.

Example 21 (the components, worked out at $k = 3, 4$)

At $k = 3$: $I_3 = P_2 \cap P_3$, P_2 (block pattern $\{1, 2\}$) generated by I_2's three generators from Example 4, and P_3 (block pattern $\{1, 2, 3\}$) exactly the Y_2-ideal displayed above. **At $k = 4$**: P_2, P_3 inherited verbatim from $k = 3$, $\overline{v}_4$ left free; P_4 is the $k = 4$ row of the Y_2-defining-ideal example above.

6. Cohen-Macaulayness

Proposition 22 (Eagon-Northcott / Hilbert-Burch, applied)

Let M be a 2×3 matrix over a polynomial ring R and J the ideal of its 2×2 minors. If J has codimension 2, then R/J is Cohen-Macaulay.

Proof of Proposition 22. Bruns and Vetter [1], Theorem (2.7), p. 13 – unchanged from the companion article's use of the same theorem, applying it here to the matrix $\begin{pmatrix} D(\overline{w}_{k-1}) \\ \overline{v}_k \end{pmatrix}$ instead of $\begin{pmatrix} \tau(\overline{w}_{k-1}) \\ \overline{v}_k \end{pmatrix}$. □

Theorem 23 (R_k/I_k is Cohen-Macaulay)

For every $k \geq 2$: R_k/I_k is Cohen-Macaulay.

Proof of Theorem 23. By Theorem 16, I_k has the expected codimension 2 for Theorem 5's presenting matrix; Proposition 22 applies directly. □

7. Radicality

Cohen-Macaulayness and §5′s component structure combine via Serre's criterion to give radicality, exactly as in the companion article's §7; the reduction-via-Serre lemma and the two Jacobian-rank lemmas (the Jacobian-rank-bound statements) are pure commutative algebra, independent of the encoding, and are not repeated here – only the Jacobian **recursion** itself needs D in place of τ.

Lemma 24 (the Jacobian's own recursion)

Write J_k for the $3 \times 3k$ Jacobian of $\overline{w}_k$ and $[u]_\times$ for the skew matrix with $[u]_\times v = u \times v$. Then $J_2 = \left[-[\overline{v}_2]_\times \;\middle|\; [\overline{v}_1]_\times\right]$, and for $k \geq 3$,

$$J_k = \left[-[\overline{v}_k]_\times DJ_{k-1} \;\middle|\; [D(\overline{w}_{k-1})]_\times\right]. \tag{23}$$

Proof of Lemma 24. Identical derivation to the companion article's own Lemma 24 with D in place of T (the matrix of τ): $\overline{w}_k = D(\overline{w}_{k-1}) \times \overline{v}_k = [D(\overline{w}_{k-1})]_\times \overline{v}_k$ is linear in $\overline{v}_k$ with $\overline{v}_k$-free coefficient matrix, giving the second block; and $\partial_t \overline{w}_k = (D\partial_t \overline{w}_{k-1}) \times \overline{v}_k = -[\overline{v}_k]_\times D\partial_t \overline{w}_{k-1}$ for t among $\overline{v}_1, \ldots, \overline{v}_{k-1}$, using $a \times b = -[b]_\times a$, gives the first. □

Example 25 (J_3, explicitly)

With variables ordered $v_{1,1}, \ldots, v_{3,3}$, the 3×9 Jacobian of $\overline{w}_3$, in the two blocks of Lemma 24′s formula – computed directly (*code/compute_signed_encoding_examples.sage*), **every entry a ± 1-coefficient binomial or monomial**, in contrast with the companion article's own J_3 example, which has several entries with explicit factors of 2. First, $-[\overline{v}_3]_\times DJ_2$ (3×6):

$$\begin{pmatrix} -v_{2,2}v_{3,2} + v_{2,3}v_{3,3} & v_{2,1}v_{3,2} & -v_{2,1}v_{3,3} & v_{1,2}v_{3,2} - v_{1,3}v_{3,3} & -v_{1,1}v_{3,2} & v_{1,1}v_{3,3} \\ v_{2,2}v_{3,1} & -v_{2,1}v_{3,1} + v_{2,3}v_{3,3} & -v_{2,2}v_{3,3} & -v_{1,2}v_{3,1} & v_{1,1}v_{3,1} - v_{1,3}v_{3,3} & v_{1,2}v_{3,3} \\ -v_{2,3}v_{3,1} & -v_{2,3}v_{3,2} & v_{2,1}v_{3,1} + v_{2,2}v_{3,2} & v_{1,3}v_{3,1} & v_{1,3}v_{3,2} & -v_{1,1}v_{3,1} - v_{1,2}v_{3,2} \end{pmatrix} \tag{24}$$

and $[D(\overline{w}_2)]_\times$ (3×3):

$$\begin{pmatrix} 0 & v_{1,2}v_{2,1} - v_{1,1}v_{2,2} & -v_{1,3}v_{2,1} + v_{1,1}v_{2,3} \\ -v_{1,2}v_{2,1} + v_{1,1}v_{2,2} & 0 & v_{1,3}v_{2,2} - v_{1,2}v_{2,3} \\ v_{1,3}v_{2,1} - v_{1,1}v_{2,3} & -v_{1,3}v_{2,2} + v_{1,2}v_{2,3} & 0 \end{pmatrix} \tag{25}$$

Lemma 26 (rank-2 witness on the new component)

For every $k \geq 2$, there is a point $x = (p, 0)$, p the Lemma 7 witness at level $k-1$, with $x \in V(I_k)$ and rank $J_{k(x)} = 2$.

Proof of Lemma 26. $J_{k(p,0)} = \left[0 \;\middle|\; [D(\overline{w}_{k-1}(p))]_\times\right]$ since $-[0]_\times DJ_{k-1}(p) = 0$, and $D(\overline{w}_{k-1}(p)) \neq 0$ since $\overline{w}_{k-1}(p) \neq 0$ (Lemma 7) and D is invertible. For $u \neq 0$, $\ker[u]_\times = \mathbb{F}u$, so $[D(\overline{w}_{k-1}(p))]_\times$ has rank 2. □

Example 27 (the rank-2 witness, at $k = 3$)

Taking $p = (\overline{v}_1, \overline{v}_2) = ((1,0,0),(0,1,0))$ (the same point used in the Y_2-point example above, where $\overline{w}_2(p) = (0,0,1) \neq 0$), $x = (p, 0) \in V(I_3)$ and $J_3(x) = [0 \mid [(0,0,1)]_\times]$, which has rank 2 directly from the explicit $[D(\overline{w}_2)]_\times$ block of the J_3 example above, evaluated at p.

Lemma 28 (rank-2 witness on an inherited component)

For every $k \geq 3$ and every component Z of $V(I_{k-1})$ with a rank-2 witness $q \in Z$, there is $x = (q, e_i) \in Z \times \mathbb{A}^3 \subset V(I_k)$ with rank $J_{k(x)} = 2$.

Proof of Lemma 28. Identical to the companion article's own proof with D in place of T: $W := D \cdot \operatorname{im} J_{k-1}(q)$ is 2-dimensional (D invertible, rank $J_{k-1}(q) = 2$); since $W \neq \mathbb{F}^3$ some $e_i \notin W$; $J_{k(q,e_i)} = \left[-[e_i]_\times DJ_{k-1}(q) \;\middle|\; 0\right]$ has rank $\dim\left([e_i]_\times W\right) = 2$ since $\ker[e_i]_\times = \mathbb{F}e_i$ meets W trivially. □

Theorem 29 (I_k is radical)

For every $k \geq 2$: I_k is radical.

Proof of Theorem 29. Induction on k, mirroring Theorem 19: Lemma 26 and Lemma 28 exhibit a rank-2-Jacobian point on every one of the $k-1$ components, giving R_0; Cohen-Macaulayness (Theorem 23) gives S_1 via Serre's criterion (the reduction-via-Serre step is pure commutative algebra, identical to the companion article's own reduction-via-Serre lemma, not repeated here). □

Verified independently in Sage at $k = 2, 3, 4$: the Jacobian recursion checked as a polynomial identity (*code/compute_signed_encoding_examples.sage* builds J_3 directly from $\overline{w}_3$'s own Jacobian, not from the recursive formula, giving an independent cross-check).

8. The Primary Decomposition Theorem

Theorem 30 (Primary Decomposition Theorem)

For every $k \geq 2$, over any field of characteristic $\neq 2$: I_k has exactly $k-1$ irreducible, radical, primary components $P_2, \ldots, P_k$, and $I_k = P_2 \cap \cdots \cap P_k$ is an irredundant primary decomposition.

Proof of Theorem 30. Identical to the companion article's own proof: Theorem 29 gives $I_k = \sqrt{I_k} = \cap_{P \in \operatorname{Min}(I_k)} P$; Theorem 19 identifies $\operatorname{Min}(I_k) = \{P_2, \ldots, P_k\}$; irredundancy is Theorem 19′s own no-containment conclusion. □

9. Summary

For every $k \geq 2$, over any field $\mathbb{F}$ of characteristic $\neq 2$: I_k has codimension 2 (Theorem 16), presented as the ideal of 2×2 minors of an explicit 2×3 matrix (Theorem 5); exactly 3 minimal generators (Theorem 10); R_k/I_k Cohen-Macaulay (Theorem 23); I_k radical (Theorem 29); and $V(I_k)$ has exactly $k-1$ irreducible components assembling into the irredundant Primary Decomposition Theorem (Theorem 30). Every statement is identical to the companion article's; what differs is the route – the signed encoding (Theorem 1) replaces every occurrence of the coordinate-swap-and-doubling map τ with the sign-diagonal involution D, removing the swap entirely and reducing every displayed formula in this article to ± 1 coefficients throughout, traced (Remark 2) to D being, up to a constant, the matrix of $\mathfrak{sl}_2$'s own Killing form in these coordinates – an explanation for the twist's existence that the companion article's own τ does not have.

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